Laboratory Manual

Module Number : 1

Module Name : Mechatronics

Sub Module Name : Electrical Engineering

Name	
Register Number	

List of Experiments

1. Verification of Basic Laws
2. Study of Protective Devices
3. Polarity Test of Single-Phase Transformer
4. Predetermination of Efficiency of Single-Phase Transformer
5. Measurement of Power in RLC Circuits
6. Load Characteristics of DC Shunt Motor
7. Study of Stepper Motor

S.No	Date	Title	Marks	Signature

EXPERIMENT – 1 (a)

Verification of Ohm's Law

Aim:

To study and verify Ohm's Law

Components Required:

Circuit Diagram:

Observation Table :

Voltage (V)	Current (mA) (Measured)	Current (mA) (Calculated)

Calculation :

Result:

EXPERIMENT – 1 (b)

Verification of Kirchoff's Laws

Aim:

To study and verify Kirchoff's voltage and current law

Components Required:

Circuit Diagram:

Kirchoff's Voltage Law

Observation Table :

Supply Voltage (V)	V1 (V)	V2 (V)

Circuit Diagram:

Kirchoff's Current Law

Observation Table :

I (mA)	I1 (mA)	I2 (mA)

Calculation :

Result :

EXPERIMENT – 2

Study of Protective Devices

Aim:

Components Required:

Circuit Diagram:

Inference and Result :

EXPERIMENT – 3

Polarity Test on Single-Phase Transformer

Aim:

Components Required:

Circuit Diagram:

Additive Polarity :

Observation Table :

V1 (V)	V2 (V)	V3= V1+V2 (V)

Subtractive Polarity :

Observation Table :

V1 (V)	V2 (V)	V3= V1-V2 (V)

Result:

EXPERIMENT – 4

Predetermination of efficiency of Single-Phase Transformer

Aim:

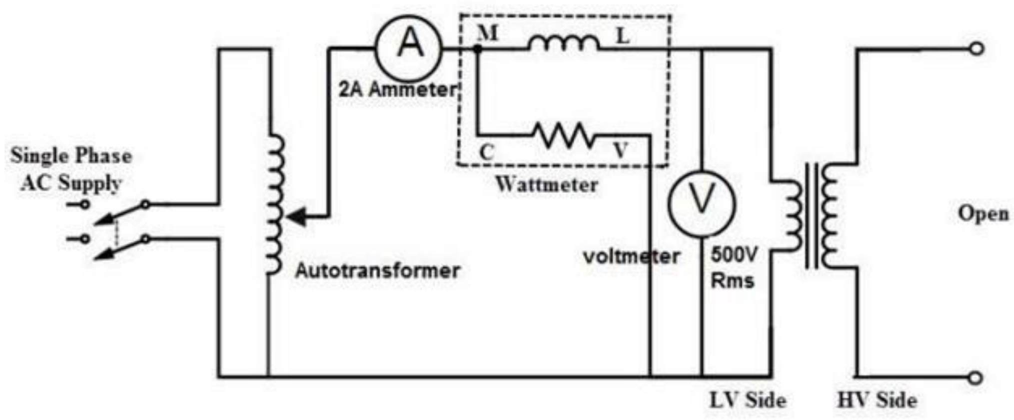
(i) To perform the open circuit test on a single-phase transformer for finding shunt parameters and core loss

(ii) To perform the short circuit test on a single-phase transformer for finding series parameters and copper loss

Components Required :

Circuit Diagram:

Open Circuit Test



Observation Table:

Type	V_o (V)	I_o (A)	W_o (W)	Power factor ($Cos\varphi$)
Core				
Shell				
Toroidal				

Sample Calculations:

The total iron loss = Wo

No load power factor = Wo / VoIo

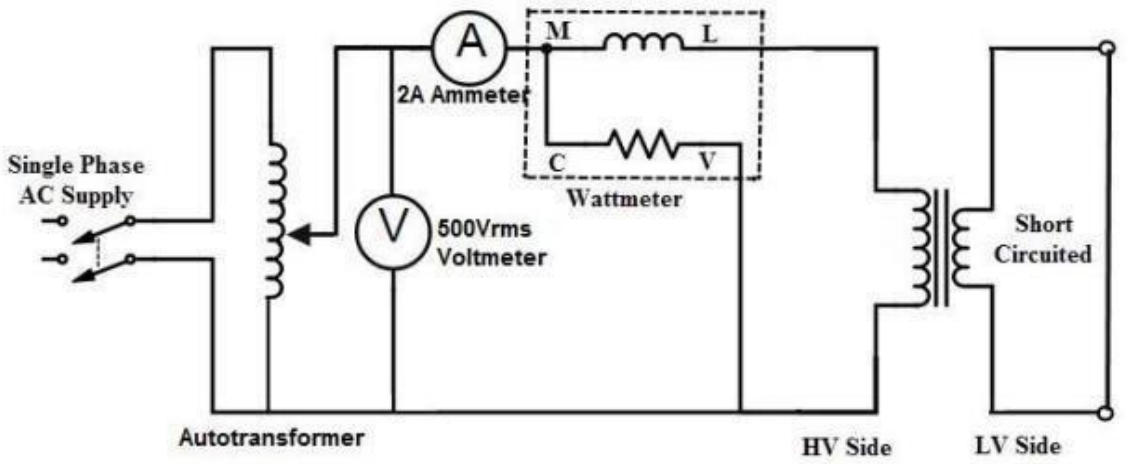
Core loss component of the current = Iw=IoCosφ
Magnetising component of current=Iu=IoSinφ

Ro=Vo/Iw

Xo=Vo/Iu

Circuit Diagram:

Short Circuit Test



Observation Table:

Type	V_{sc} (V)	I_{sc} (A)	W_{sc} (W)	Power factor (Cosφ)
Core				
Shell				
Toroidal				

Sample Calculations:

Total copper loss = Wsc

$R01 = Wsc / Isc^2$

$Z01 = Vsc / Isc$

$X01 = \sqrt{(Z01^2 - R01^2)}$

Equivalent Circuit:

Result:

<u>EXPERIMENT – 5</u>

Measurement of Power in RLC Circuit

<u>Aim:</u>

<u>Components Required</u> :

<u>Circuit Diagram:</u>

<u>Series RLC Circuit</u>

Parallel RLC Circuit

Observation Table:

S.No	Type of Connection	Load Voltage (V)	Load Current (A)	Active Power (kW)	Reactive Power (kVAR)	Apparent Power (kVA)
1.	Series					
2.	Parallel					

Calculation:

Result:

EXPERIMENT - 6

Load Characteristics of DC Shunt Motor

Aim:

Components Required :

Circuit Diagram:

Calculation:

Torque is calculated by using the formula T = F * r N-m

Where,

F= Force (Load) & r= Radius of motor frame.

Force can be calculated by observing the load applied.

F = Load weight * 9.81 N-m

Observation Table:

Sl.No.	Load Voltage (V)	Load Current (A)	Speed (RPM)	Force= F(Nm)	Torque= F*r (N-m)

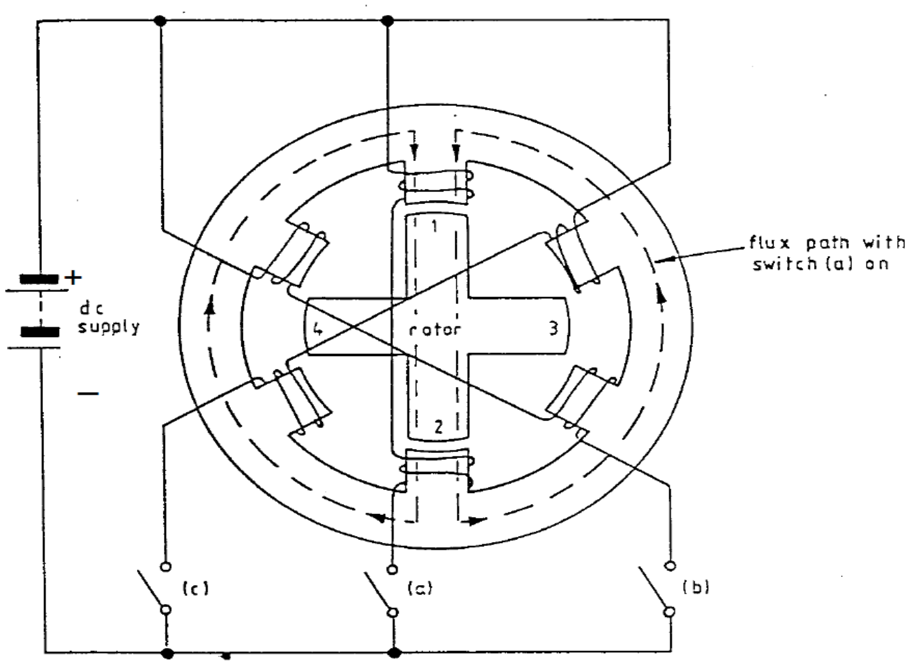
Result:

EXPERIMENT - 7

Study of Stepper Motor

Aim:

Circuit Diagram:



A stepping motor has a number of windings, or groups of windings, which can be separately energised. Each is called a 'phase' of the motor. The number of phases is typically two, three or four.

If one phase is energised, the motor's shaft will be held still. In order to move the shaft, a supply is switched sequentially to each phase in turn. Each phase, when it is energised, pulls the shaft into a new position, so that the motor moves in a series of 'steps'. The construction is such that when the switching cycle is complete and another similar cycle is begun, the first step in the new cycle will continue the stepping motion.

In the case of a reluctance-type stepping motor, the rotor aligns itself to a position giving the minimum reluctance of the magnetic path which is energised at the time. Figure A3-1 is a diagram of the three-phase reluctance-type stepping motor about to be constructed. If switch (a) is on, the rotor will align itself as shown, giving the easiest possible path for the magnetic flux. If now (a) is switched off and (b) is switched on, the rotor will move through 30° clockwise, to align rotor teeth 3 and 4 with the stator teeth carrying the (b) phase windings.

The step angle can be calculated from the number of stator slots, S, and number of rotor slots, R. The number of steps per revolution is

$$N = \frac{SR}{S - R}$$

The step angle is $\dfrac{360^\circ}{N}$

Energised One Phase (clockwise rotation)			
Pointer Position	**Switch Setting**		
	a	**b**	**c**
1	on	off	off
2	off	on	off
3	off	off	on
4	on	off	off
5	off	on	off
6	off	off	on
7	on	off	off
8	off	on	off
9	off	off	on
10	on	off	off
11	off	on	off
12	off	off	on

Energised One Phase (anti-clockwise rotation)			
Pointer Position	**Switch Setting**		
	a	**b**	**c**
12	off	off	on
11	off	on	off
10	on	off	off
9	off	off	on
8	off	on	off
7	on	off	off
6	off	off	on
5	off	on	off
4	on	off	off
3	off	off	on
2	off	on	off
1	on	off	off

Energised Two Phase (clockwise rotation)			
Pointer Position	Switch Setting		
	a	b	c
1	on	on	off
2	off	on	on
3	on	off	on
4	on	on	off
5	off	on	on
6	on	off	on
7	on	on	off
8	off	on	on
9	on	off	on
10	on	on	off
11	off	on	on
12	on	off	on

Energised Two Phase (anti-clockwise rotation)			
Pointer Position	Switch Setting		
	a	b	c
12	on	off	on
11	off	on	on
10	on	on	off
9	on	off	on
8	off	on	on
7	on	on	off
6	on	off	on
5	off	on	on
4	on	on	off
3	on	off	on
2	off	on	on
1	on	on	off

Result: